MSCC on Raspberry Pi — one how-to

Pi: 4 or 5, **64-bit Raspberry Pi OS** (desktop recommended).

Packages: use files in `pi-install/packages/` (this folder).

Updated: 2026-09 — servers **1.0.40**, init-gui **1.0.12**, portaudio **19.8.2**, UI **0.6.36**.

If a filename in `packages/` differs, **use the real name on disk**.

Big picture

Role	Where it runs
Servers (`ms-sdr`, `sdrcore-recv`, `sdrcore-trans`)	**Always on the Pi** (next to the radio)
Operate UI	**Windows WPF** on a PC, and/or **Avalonia MSCC UI** on the Pi
Digi apps (WSJT, etc.)	Prefer **on the Pi** (VirtualA / VirtualB)
Remote operator voice	Optional: phones AF + mic on a **Windows PC** via **MsccRemotePhones**

```
Windows WPF client   ■■UDP■■■   Pi servers   ■■USB■■■   Proficio / Geminus
        ▲                          ■
        ■                          ■■■ digi: VirtualA/B (local)
        ■■■ optional remote AF ■■■■
            MsccRemotePhones
            RX 9100 / TX 9101
```

Typical setups

1. **Headless Pi + Windows WPF** — Pi runs servers only; you operate from the PC.
2. **Pi with Avalonia UI** — servers + **MSCC UI** on the Pi (local `127.0.0.1`).
3. **Remote voice** — digi on Pi; operator phones/mic on Windows (**Remote Audio**).

What you need

1. Pi on the network; desktop Terminal for install.
2. Contents of `pi-install/packages/` on the Pi (`~/Downloads` or USB).
3. Proficio/Geminus powered when configuring (recommended).
4. Operator audio plan (especially **Pi 4**): prefer **I²S HAT** or digi-only — avoid USB headset + Proficio USB on Pi 4 under load.

A. Install servers (required)

On the Pi Desktop Terminal, `cd` to the folder with the `.deb` files:

```
cd ~/Downloads/packages    # or wherever you copied pi-install/packages
```

1) PortAudio (first)

```
sudo apt install -y ./mscc-portaudio_19.8.2_arm64.deb
ldconfig -p | grep portaudio
# expect: /usr/local/lib/libportaudio.so.2
```

2) Main MSCC package (servers + firmware upload tools)

```
sudo apt update
sudo apt install -y ./mscc_1.0.40_arm64.deb
```

Optional helper (if present):

```
chmod +x ../install-mscc.sh   # if you copied install-mscc.sh next to packages
./install-mscc.sh
```

This installs:

Piece	Notes		
Servers	`$HOME/mscc/` — `ms-sdr`, `sdrcore-recv`, `sdrcore-trans`		
Firmware Upload	`bootloader` + `bootloader-gui` — menu **MSCC** → **Firmware Upload**		
Start / Stop / Status	Desktop **MSCC** menu + `mscc start\	stop\	status`
Digi sinks	VirtualA / VirtualB		
Config seed	`$HOME/.local/mscc/` if empty		

3) Setup wizard

```
sudo apt install -y ./mscc-init-gui_1.0.12_all.deb
```

4) Log out / log in once (first install)

Needed so `dialout` / `audio` / `plugdev` groups apply (CAT and devices).

Check digi sinks:

```
pactl list short sinks | grep Virtual
# if empty:
systemctl --user enable --now mscc-virtual-audio
```

5) Configure once — MSCC Init

Menu **MSCC** → **MSCC Init** (or `mscc-init`):

- Keyer, CAT / PTT
- Operator speaker & mic (not Proficio I/Q devices)
- Digi fixed to VirtualA / VirtualB
- **Proficio MKII vs Legacy** (`PROFICIO-MKII`) — use **Legacy (0)** for semi-break-in / external keyer radios
- Optional: start servers at end of wizard

Config lives in `$HOME/.local/mscc/` (upgrades do **not** overwrite).

6) Everyday servers

Menu	Action
MSCC Start	Start servers
MSCC Stop	Stop servers
MSCC Status	Health check
MSCC Init	Reconfigure
Firmware Upload	Flash Proficio/Geminus `.cyacd` (BOOT jumper → LOADER)

```
mscc start
mscc status
```

```
mscc stop
```

B. Install Avalonia UI on the Pi (optional)

Use when you want the operate GUI **on the Pi** (or a second Linux box).

```
sudo apt install -y ./mscc-ui_0.6.36_arm64.deb
```

Item	Detail
Run	Menu **MSCC UI**, or `mscc-ui`
Default host	`127.0.0.1` port 8888 (start servers first)
Settings	`~/.config/MSCC/mscc-avalonia.ini` (survive reinstall)
Uninstall	`sudo apt remove mscc-ui` (does not remove servers)

UI features to know

Feature	Where / notes
Remote Audio	Left rail checkbox — with **Phones** selected, sends audio device **2** (remote mic). Greyed on **Digital**. Sticky in ini.
External electronic keyer (legacy)	CW tab — sets `PROFICIO-MKII=0` in host `mscc.ini`; restart servers to apply.
Connect	Host / ports for remote Pi if UI is not on the radio machine

Note: Rebuild the UI `.deb` after client code changes (`Avalonia-Migration/build-mscc-ui-deb.ps1`), then refresh `pi-install/packages/` with `collect-packages.ps1`.

More detail: `mscc-ui/Avalonia-Migration/INSTALL-MSCC-UI.md`.

C. Windows client (WPF) against the Pi

1. Pi: **MSCC Start**; note Pi IP (`hostname -I`).
2. Windows: run **MSCC.Wpf** from `C:\mscc-net9` (or your install).
3. Settings: Server IP = Pi; **Launch Servers** usually **off** (servers already on Pi).
4. **Start** / Connect.

Legacy CW: check **Use external electronic keyer** on CW tab, or set `PROFICIO-MKII=0` on the Pi (`mscc.ini` / Init / `MSCC-Remote` if used on Windows host).

D. Remote audio (operator voice on Windows, digi on Pi)

Servers already support it. You need the companion app + UI checkbox.

On the Pi

Edit (or create) phones stream target — Windows PC IP:

`$HOME/.local/mscc/remote-phones.ini`

```
ENABLED=1
HOST=192.168.x.x
PORT=9100
```

Mic listen port (usually default is enough):

`$HOME/.local/mscc/remote-mic.ini`

```
PORT=9101
```

Restart servers after editing. Digi stays on VirtualA/B.

On Windows

1. Build/run **MsccRemotePhones** from `mscc-remote-audio/` (see that folder's README).
2. **RX:** listen **9100** (phones AF from Pi).
3. **TX:** host = **Pi IP**, port **9101**, start mic TX.
4. In **WPF or Avalonia**: Audio = **Phones**, check **Remote Audio** → client sends `0x9B` **/ data 2**.
5. Digital path: leave **Digital** selected for digi; Remote Audio is ignored (always device **0**).

Full handoff: `mscc-remote-audio/STEW-REMOTE-AUDIO.md`.

Test without UI (live session):

```
cd ...\mscc-remote-audio
.\Set-AudioDevice.ps1 -HostName PI_HOSTNAME -Mode remote
```

E. Firmware upload (bootloader)

Included in the `mscc` package — no separate `.deb`.

1. Power off radio → install **BOOT** jumper → power on → Morse **LOADER** / USB `04b4:b71d`.
2. **Stop** MSCC servers.
3. Pi menu **MSCC** → **Firmware Upload** (`bootloader-gui`) or CLI `bootloader /path/to/file.cyacd`.
4. Use the correct tree's `.cyacd` (Proficio/Geminus MKII or Legacy under `Release-*`).
5. Power off → **remove BOOT jumper** → power on → Proficio `16c0:05dc`.

Details: `Release-Proficio-Legacy/STEW-FIRMWARE-UPDATE.md` (same procedure for MKII trees).

F. Updating packages later

1. On the Windows/dev PC, put new builds in the usual trees (`mscc-deb/`, `mscc-ui/Avalonia-Migration/`, etc.).
2. Refresh the kit:

```
cd C:\Users\n8vet\OneDrive\Documents\GitHub\mscc-station
powershell -NoProfile -ExecutionPolicy Bypass -File .\pi-install\collect-packages.ps1
```

3. Copy new files from `pi-install/packages/` to the Pi.
4. On the Pi:

```
cd ~/Downloads/packages
sudo apt install -y ./mscc-portaudio_….deb ./mscc_….deb ./mscc-init-gui_….deb ./mscc-ui_….deb
```

`apt install ./file.deb` upgrades in place. **Config under** `~/.local/mscc/` **and UI** `~/.config/MSCC/` **is kept.**

Then:

```
mscc restart
# or MSCC Stop / Start from the menu
```

Update the version table at the top of this file when you change the kit.

Quick reference — where things live in the repo

Need	Path
This kit + how-to	`pi-install/`
Server packaging / older install prose	`mscc-deb/`
Init GUI sources	`mscc-init-gui/`
Avalonia UI + build script	`mscc-ui/Avalonia-Migration/`
Windows WPF	`mscc-ui/windows-work-tree/`
Remote phones app	`mscc-remote-audio/`
Radio `.cyacd`	`Release-Proficio-*`, `Release-Geminus-*`

Checklist (first Pi)

- [ ] PortAudio → main `mscc` → init-gui → log out/in
- [ ] VirtualA/B present; **MSCC Init** done (`PROFICIO-MKII` correct)
- [ ] **MSCC Start**; Status OK; Proficio USB seen
- [ ] Optional: **mscc-ui** on Pi, or Windows WPF to Pi IP
- [ ] Optional: Remote Audio + MsccRemotePhones
- [ ] Optional: Firmware Upload with correct `.cyacd`